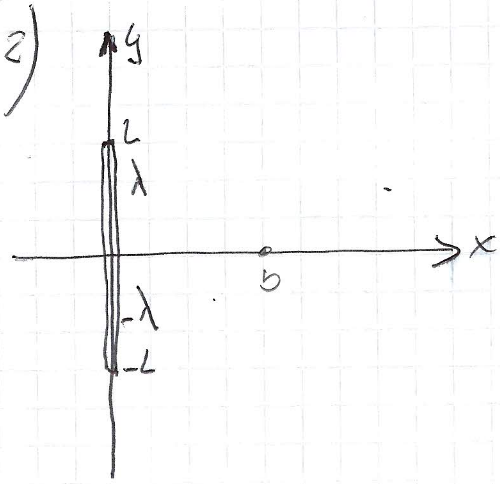
2)



Calculo el campo electrico generado por la parte superior

$$E_s = \frac{1}{4\pi\varepsilon_0}\int_{V_r} \frac{q\,(\bar{r}-\bar{r}')}{\|\bar{r}-\bar{r}'\|^3}\, dr$$

$\bar{r}$ punto de campo $(\cancel{b},0)$ $\quad dq_s = \lambda\, d\cancel{x}y$

$\bar{r}'$ punto de fuente $(0,y)$

$$(\bar{r}-\bar{r}') = (b,-y)$$

$$\|\bar{r}-\bar{r}'\| = \sqrt{b^2+y^2}$$

$$\|\bar{r}-\bar{r}'\|^3 = (b^2+y^2)^{3/2}$$

$$E_s = \frac{1}{4\pi\varepsilon_0}\int_0^L \lambda\, dy\, \frac{(\cancel{b},-y)}{(\cancel{b}^2+y^2)^{3/2}}$$

$$E_{sx} = \frac{\lambda}{4\pi\varepsilon_0}\int_0^L \frac{\cancel{b}}{(\cancel{b}^2+y^2)^{3/2}}\, dy$$

integral 96 de tabla

$$E_{sx} = \frac{\lambda b}{4\pi\varepsilon_0}\int_0^L (\cancel{b}^2+y^2)^{-3/2}\, dy \Rightarrow E_{sx} = \frac{\lambda \cancel{b}}{4\pi\varepsilon_0}\, \frac{y}{\cancel{b}^2\sqrt{y^2+\cancel{b}^2}}\Bigg|_0^L$$

$$E_{sx} = \frac{\lambda}{4\pi\varepsilon_0}\, \frac{L}{\cancel{b}\sqrt{L^2+\cancel{b}^2}}$$

integral 197

$$E_{sy} = \frac{\lambda}{4\pi\varepsilon_0}\int_0^L \frac{-y}{(\cancel{b}^2+y^2)^{3/2}} \Rightarrow E_{sy} = \frac{\lambda}{4\pi\varepsilon_0}\, \frac{1}{\sqrt{y^2+\cancel{b}^2}}\Bigg|_0^L$$

$$E_{sy} = \frac{\lambda}{4\pi\varepsilon_0}\left[\frac{1}{\sqrt{L^2+\cancel{b}^2}} - \frac{1}{\cancel{b}}\right]$$ siempre negativo

Calculo el campo electrico generado por la parte inferior

$\bar{r}$ punto del campo $(x,0)$ $\quad q=\lambda\, dy$

$\bar{r}'$ punto de fuente $(0,y)$

$(\bar{r}-\bar{r}')=(x,-y)$

$\|\bar{r}-\bar{r}'\|^3=\sqrt{x^2+y^2}^{\,3}$

$$E_i=\frac{1}{4\pi\varepsilon_0}\int_{-L}^{0}\lambda\, dy\,\frac{(x-y)}{(x^2+y^2)^{3/2}}$$

$$E_{ix}=\frac{\lambda x}{4\pi\varepsilon_0}\int_{-L}^{0}(x^2+y^2)^{3/2}dy \overset{\text{integral 196}}{\Rightarrow} E_ix=\frac{\lambda x}{4\pi\varepsilon_0}\ \frac{y}{x^2\sqrt{y^2+x^2}}\Bigg|_{-L}^{0}$$

$$E_{ix}=\frac{\lambda}{4\pi\varepsilon_0 x}\ \frac{L}{\sqrt{L^2+x^2}}$$

$$E_{iy}=\frac{\lambda}{4\pi\varepsilon_0}\int_{-L}^{0}\frac{-y}{(\sqrt{y^2+x^2})^{3/2}} \overset{\text{integral 197}}{\Rightarrow} E_{iy}=\frac{\lambda}{4\pi\varepsilon_0}\ \frac{1}{\sqrt{y^2+x^2}}\Bigg|_{-L}^{0}$$

$$E_{iy}=\frac{\lambda}{4\pi\varepsilon_0}\left[\frac{1}{x}-\frac{1}{\sqrt{L^2+x^2}}\right]$$

Calculo el campo total en x en el punto b y en y

$$E_x(b)=E_S(x=b)+E_I(x=b)=\frac{\lambda_0}{4\pi\varepsilon_0}\ \frac{L}{b\sqrt{L^2+b^2}}+\frac{-\lambda_0}{4\pi\varepsilon_0}\ \frac{L}{b\sqrt{L^2+\ldots}}$$

$E(x=b)=0$

$$E_y(b)=E_{Sy}(x=b)+E_{iy}(x=b)=\frac{\lambda_0}{4\pi\varepsilon_0}\left[\frac{1}{\sqrt{L^2+b^2}}-\frac{1}{b}\right]+\frac{-\lambda_0}{4\pi\varepsilon_0}\left[\frac{1}{b}-\frac{1}{\sqrt{L^2\ldots}}\right.$$

$$E_y(x=b)=\frac{\lambda_0}{2\pi\varepsilon_0}\left[\frac{1}{\sqrt{L^2+b^2}}-\frac{1}{b}\right]$$ siempre negativo y decreciente

b)

$$V_{2L} - V_{3L} = \int_{3L}^{2L} E_x \, dx \Rightarrow V_{2L} - V_{3L} = \int_{3L}^{2L} 0 \, dx$$

$$V_{2L} - V_{3L} = 0 \qquad \boxed{V_{2L} = V_{3L}}$$

c)

$$V_{3L} - \underbrace{V_{\infty}}_{0} = \int_{3L}^{\infty} E(x) \, dx \Rightarrow V_{3L} = \int_{3L}^{\infty} 0 \, dx \Rightarrow V_{3L} = 0$$